

Atelectasis (Collapse)

- Loss of lung volume caused by inadequate expansion of air spaces
- Ventilation perfusion imbalance and hypoxia.
- classified into three forms

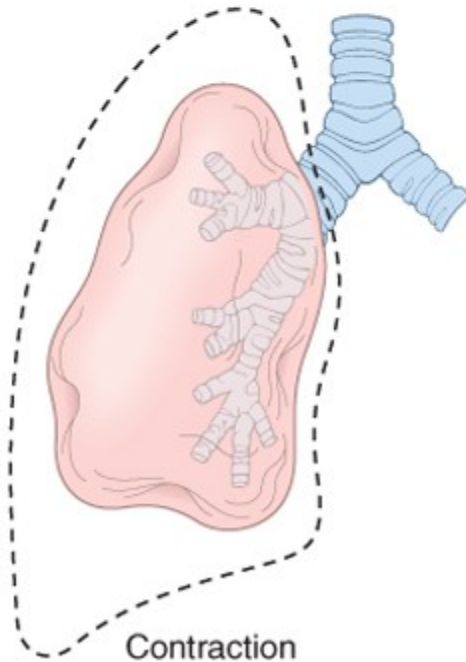
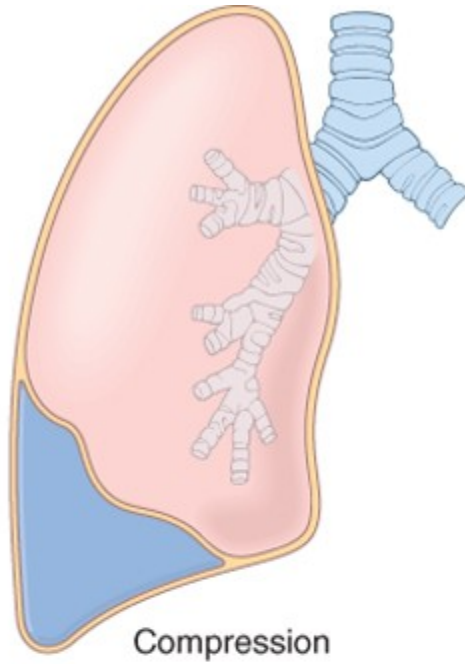
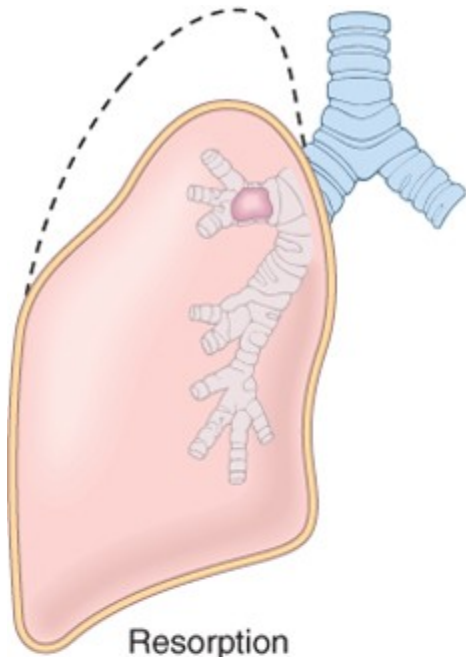
Resorption atelectasis

- Occurs when an obstruction prevents air from reaching distal airways.
- Any air present gradually becomes absorbed, and alveolar collapse follows.
- The most common cause of resorption collapse is obstruction of a bronchus.
- Resorption atelectasis most frequently occurs postoperatively due to intrabronchial mucous or mucopurulent plugs
- May also result from foreign body aspiration (particularly in children), bronchial asthma, bronchiectasis, chronic bronchitis, or intrabronchial tumor.

Compression atelectasis

- Is usually associated with accumulation of fluid, blood, or air within the pleural cavity.
- A frequent cause is pleural effusions occurring in the setting of congestive heart failure.
- Leakage of air into the pleural cavity (pneumothorax) also leads to compression atelectasis.
- Basal atelectasis resulting from a failure to breath deeply commonly occurs in bedridden patients, in patients with ascites, and during and after surgery.

- **Contraction atelectasis (or cicatrization atelectasis)**
- Occurs when local or diffuse fibrosis affecting the lung or the pleura hamper lung expansion



OBSTRUCTIVE Vs RESTRICTIVE PULMONARY DISEASES

Classification is based on pulmonary function tests

Obstructive disease (airway disease)

Characterized by an increase in resistance caused by partial or complete obstruction at any level

Decreased expiratory flow rate, usually measured by forced expiratory volume at 1 second (FEV_1)

Thus, the ratio of FEV to FVC is characteristically decreased

The major diffuse obstructive disorders

- Emphysema
- Chronic bronchitis
- Bronchiectasis
- Asthma

Restrictive disease

- Characterized by reduced expansion of lung parenchyma, with decreased total lung capacity.
- FVC is reduced and the expiratory flow rate is normal
- Restrictive diseases occur in
 - ✓ Chest disorders – kyphoscoliosis, polio, obesity, pleural disease
 - ✓ Acute or chronic interstitial & infiltrative diseases
 - ARDS, pneumoconioses, interstitial fibrosis of unknown etiology, infiltrative conditions (e.g., sarcoidosis).

Chronic obstructive pulmonary disease (COPD)

Emphysema

Emphysema is characterized by permanent enlargement of the air spaces distal to the terminal bronchioles, accompanied by destruction of their walls without significant fibrosis

Four main types

1.Centriacinar

Destruction of central portion with sparing of distal airways

Upper lobes > lower

Cause: smoking

2.Panacinar

Uniform injury

Lower lobes > upper

Cause: alpha-1-antitrypsin deficiency

3.Distal Acinar (paraseptal) Emphysema

The proximal portion of the acinus is normal, but the distal part is predominantly involved.

More striking adjacent to the pleura

Occurs adjacent to areas of fibrosis, scarring or atelectasis

More severe in the upper half of the lungs

Sometimes form cyst like structures(bullae)

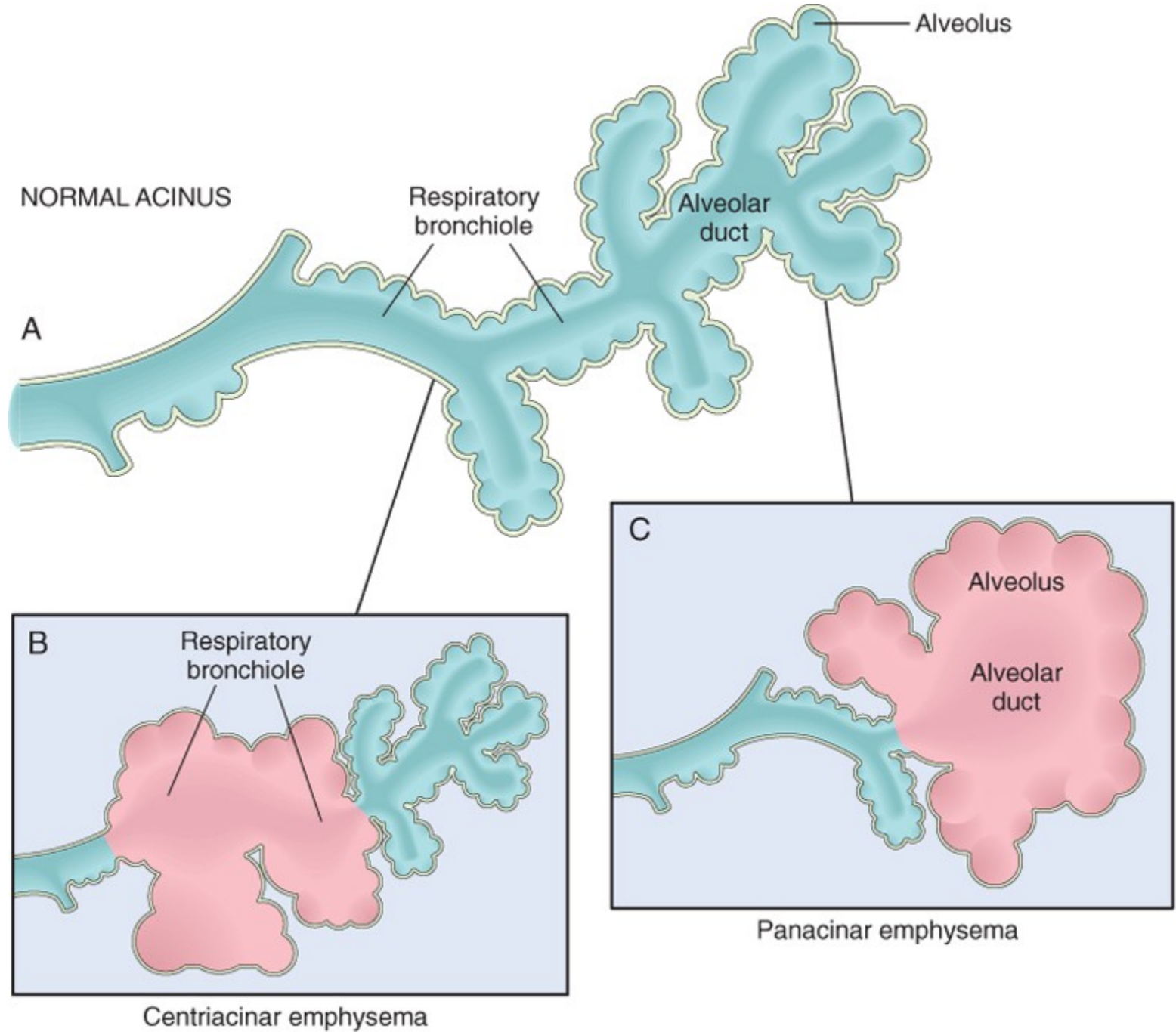
Underlies many of the spontaneous pneumothorax

4. Airspace enlargement with fibrosis (Irregular)

The acinus is irregularly involved

Associated with scarring

Asymptomatic & clinically insignificant



Pathogenesis

The pathogenesis of centriacinar and panacinar is not completely understood. Current opinion favors emphysema arising as a consequence of *two critical imbalances*:

Protease-antiprotease imbalance

Oxidant-antioxidant imbalance

COPD is characterized by mild chronic inflammation through out the airways, parenchyma, & pulmonary vasculature.

Protease – antiprotease theory

Alveolar wall destruction occurs from an imbalance between proteases (mainly elastases) & antiproteases in the lungs

Any stimulus that increases either the number of leukocytes in the lung or the release of their elastase – containing granules increases elastolytic activity

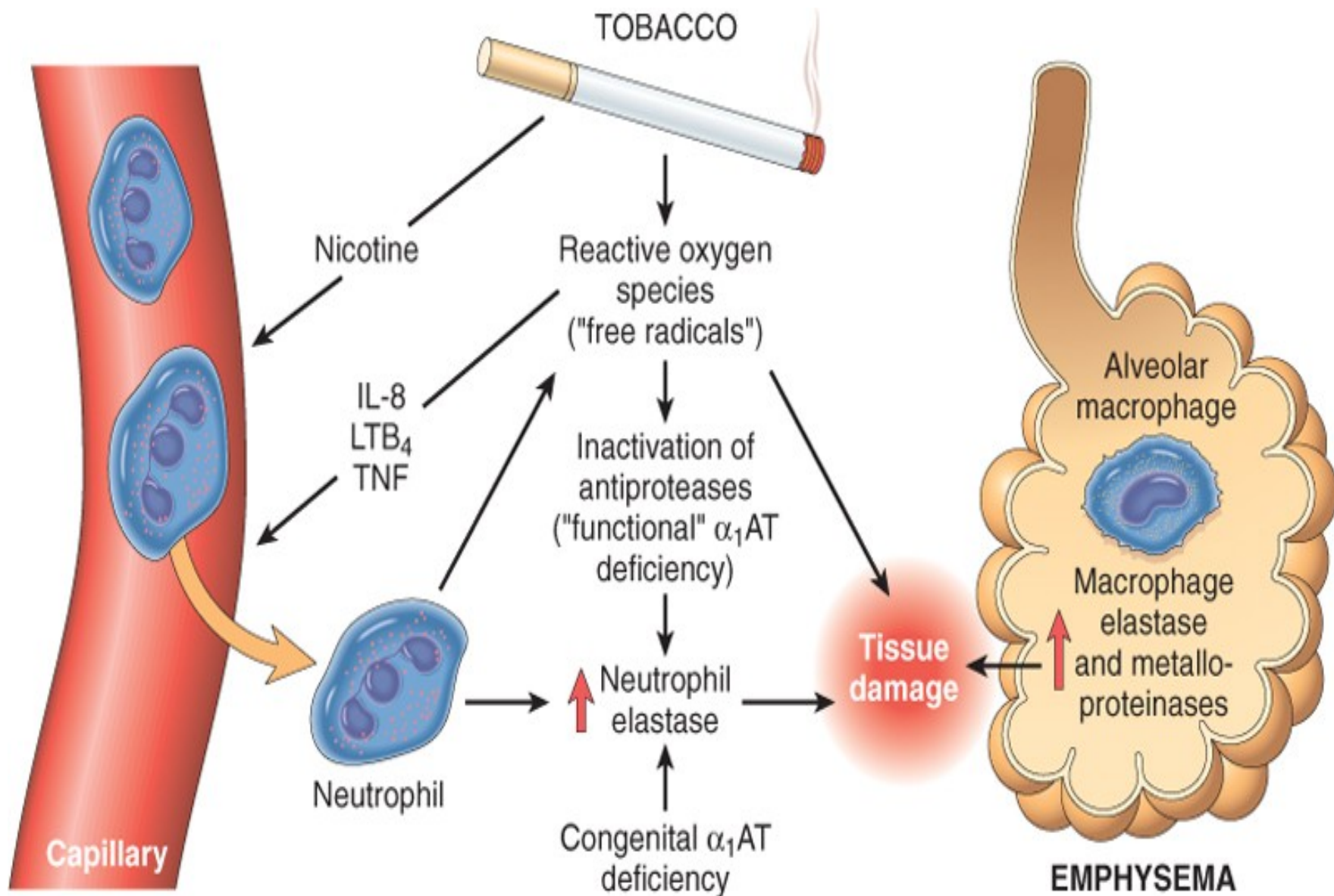
Oxidant-antioxidant imbalance

Stimulated neutrophils also release oxygen free radicals which inhibit α 1- AT activity

With low levels of serum α 1- AT the process of elastic tissue destruction is unchecked, with consequent emphysema

Ctd...

- In smokers – both increased elastase availability & decreased anti elastase activity occur
- In smokers - both neutrophils & macrophages accumulate in the alveoli
- It may involve the direct chemo attractant effect of nicotine as well as reactive oxygen species contained in smoke
- Accumulated neutrophils are activated & release their enzymes (elastase, proteinase 3, cathepsin G)
- Smoking also plays a role in perpetuating the oxidant – anti oxidant imbalance
- Oxidative injury also results in functional α 1- AT deficiency



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Clinical course

Pulmonary function tests reveal reduced FEV1 with normal or near-normal FVC.

Hence, the FEV1 to FVC ratio is reduced.

Do not appear until one third of pulmonary parenchyma is damaged

Dyspnea is first symptom & insidious

Cough or wheezing

Weight loss

Expiratory air flow limitation by spirometry

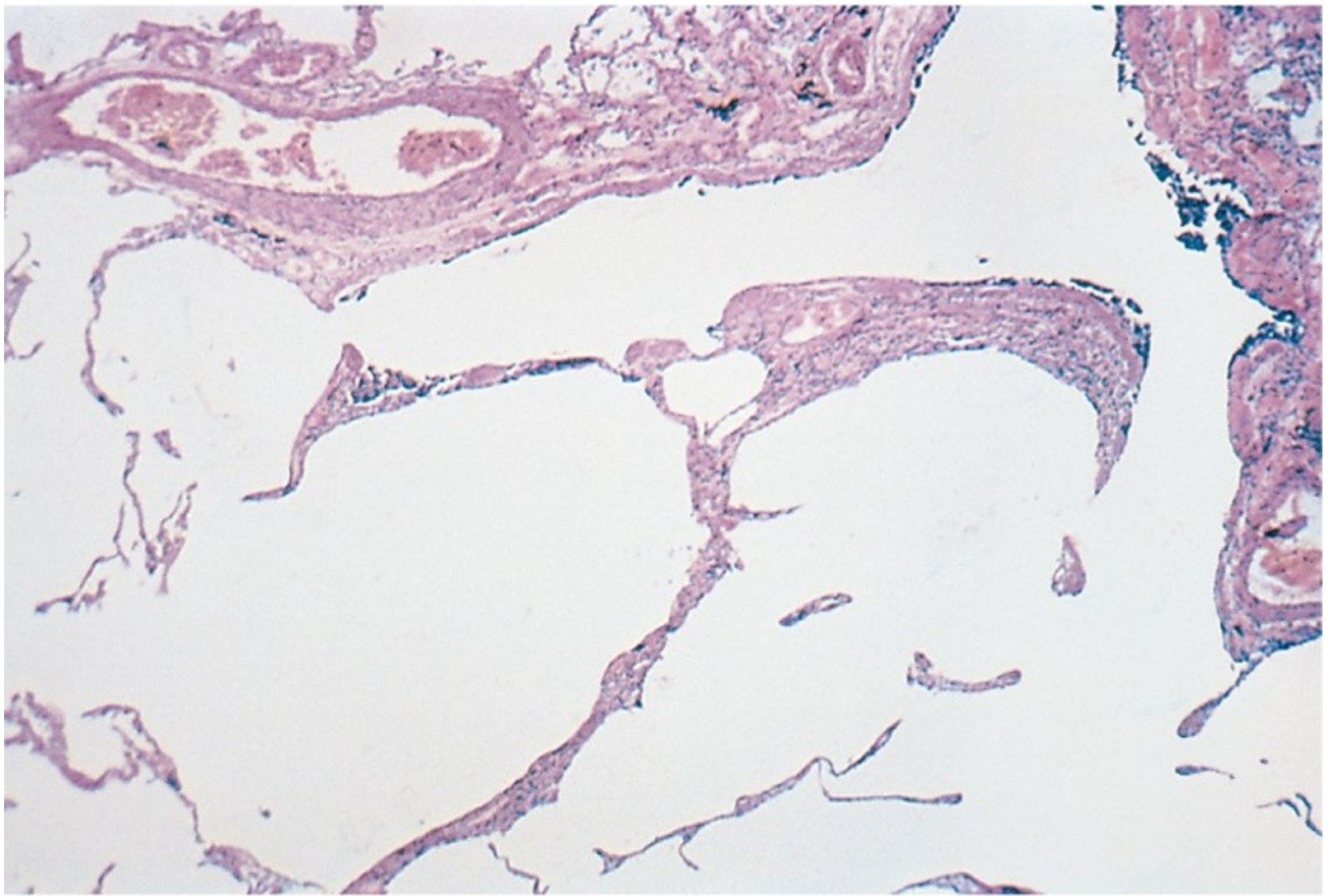
Development of cor-pulmonale & CHF secondary to pulmonary HTN

Morphology

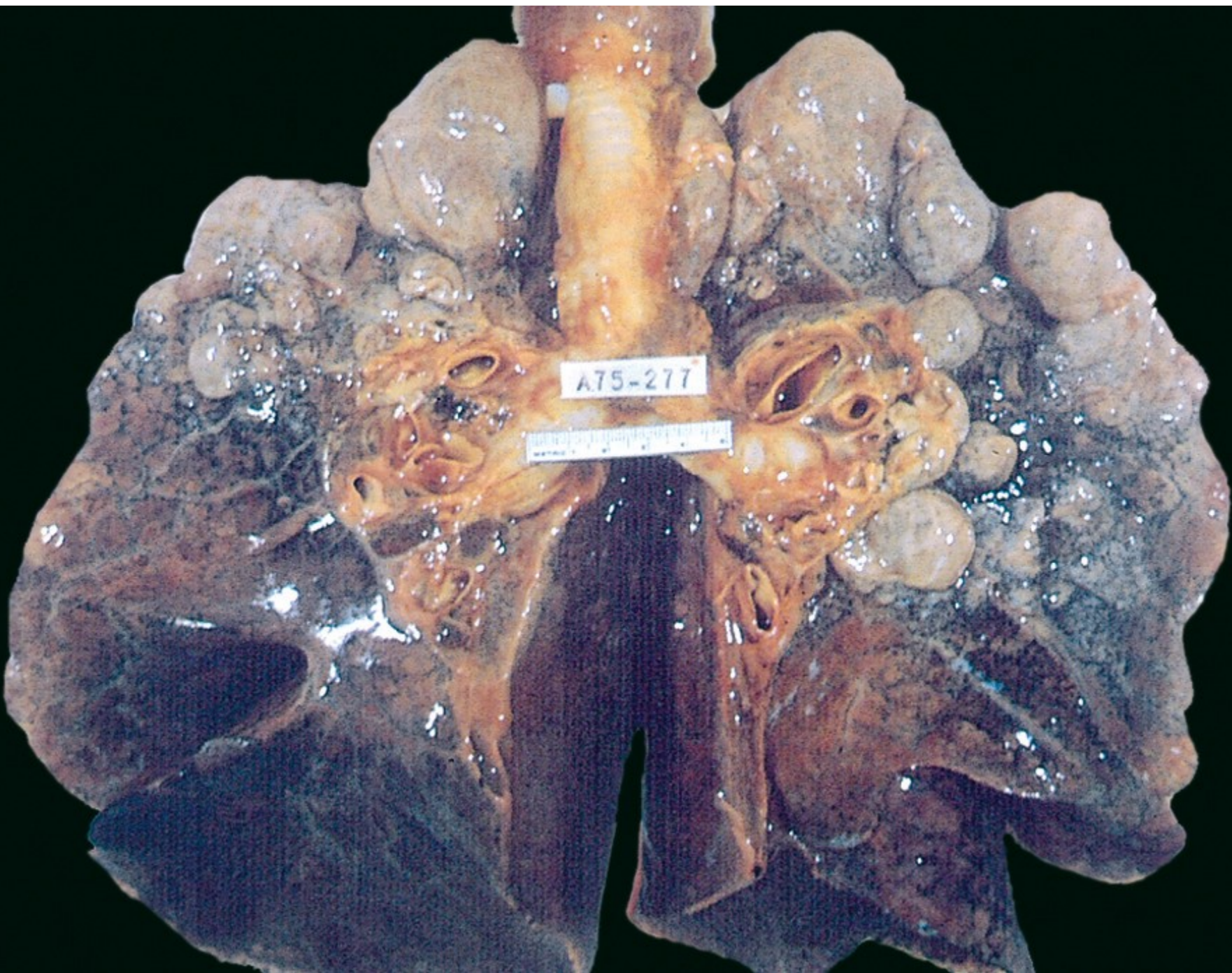
Panacinar emphysema produces voluminous lungs

Large apical blebs or bullae are characteristic of irregular emphysema & distal acinar emphysema

Respiratory bronchioles & vasculature of the lung are deformed



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Cause of death in COPD

Respiratory acidosis & coma

Rt sided heart failure

Massive collapse of lungs secondary to pneumothorax

Conditions Related to Emphysema

Compensatory hyperinflation

Obstructive hyperinflation

Bullous emphysema

Interstitial emphysema

Chronic bronchitis

Common in habitual smokers & inhabitants of smog ridden cities

When persists for years

1. progress to COPD
2. lead to heart failure
3. fertile ground for cancer

Defined as persistent cough with sputum production for at least 3 months in at least 2 consecutive yrs, in the absence of any other identifiable cause.

Pathogenesis

Chronic irritation by inhaled substances-smoking in 90%

Bacterial & viral infections are important in triggering acute exacerbation

Most frequent in middle aged male

More common in heavy smokers (4-10X)

Ctd...

Hypersecretion of mucus in the large airways associated with hypertrophy of the submucosal glands

Proteases from neutrophils & matrix metalloproteinase stimulate this hypersecretion

Role of infections appears to be secondary

Morphology

Hyperemia, swelling, edema of mucus membranes accompanied by excessive secretion

Chronic inflammation of the airways, & enlargement of mucus secreting glands

Reid index ratio i.e the thickness of the mucus gland layer to the bronchial wall

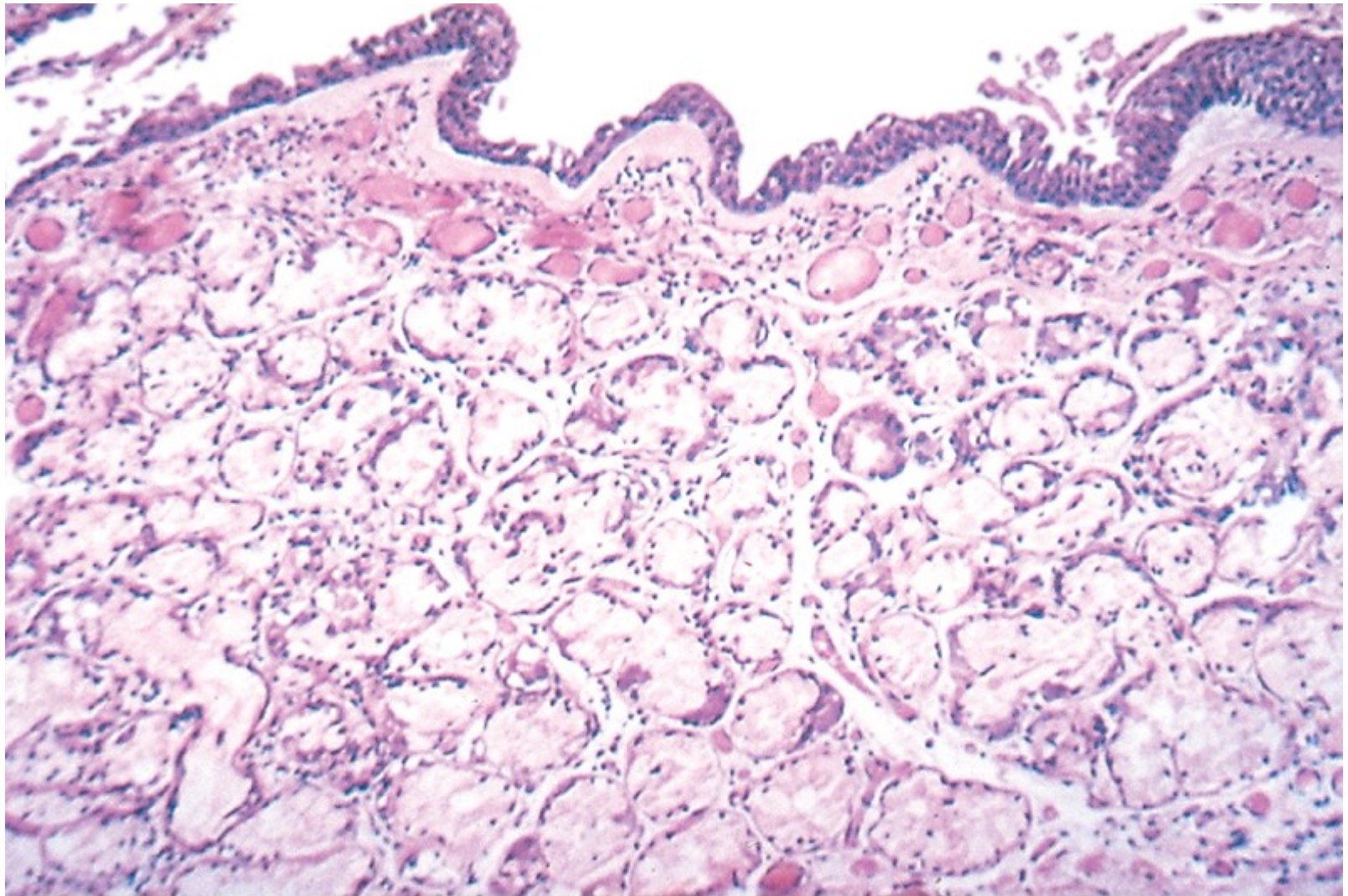
Ctd...

Reid index (normally 0.4) is increased

Bronchial epithelium may exhibit squamous metaplasia & dysplasia

Marked narrowing of bronchioles

In severe cases, bronchiolitis obliterans



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Clinical features

Persistent productive cough

Dyspnea

COPD(hypoxemia,hypercapnia,cyanosis)

Col pulmonale with cardiac failure

	Bronchitis	Emphysema
Age (yrs)	40-45	50-75
Dyspnea	Mild, late	Sever, early
Cough	Early, copious sputum	Late, scanty sputum
Infections	common	occasional
Resp. insufficiency	Common treat	Terminal
Col pulmonale	common	Rare, terminal
Airway resistance	increased	Normal or slightly increase
Elastic recoil	normal	low

Asthma

It is a chronic inflammatory disorder of the airways that causes recurrent episode of wheezing, breathlessness, chest tightness & cough particularly at night &/or in the early morning

The hallmarks of asthma are intermittent, reversible airway obstruction; chronic bronchial inflammation with eosinophils; bronchial smooth muscle cell hypertrophy and hyperreactivity; and increased mucus secretion.

Inflammation causes an increase in airway responsiveness to a variety of stimuli

Eosinophils, mast cells, macrophages, T lymphocytes, neutrophils & epithelial cells play a role

Categorization

1. depending on the frequency & severity of symptoms
 - Mild
 - Moderate
 - Severe
2. inciting agent
 - Intrinsic (non atopic)
 - Extrinsic (atopic)

Atopic Asthma

Usually begins in childhood

Triggered by env'tal pollutants

Positive family history is common

Asthmatic attacks often preceded by allergic rhinitis, urticaria or eczema

Skin test is positive to allergen injection

- The classic atopic form is associated with excessive type 2 helper T (TH2) cell activation.
- Cytokines produced by TH2 cells account for most of the features of atopic asthma — IL-4 and IL-13 stimulate IgE production, IL-5 activates eosinophils, and IL-13 also stimulates mucus production.
- IgE coats submucosal mast cells, which on exposure to allergen release their granule contents and secrete cytokines and other mediators.
- Mast cell–derived mediators produce two waves of reaction: an early (immediate) phase and a late phase
- The early-phase reaction is dominated by bronchoconstriction, increased mucus production, and vasodilation.
- Bronchoconstriction is triggered by mediators released from mast cells, including histamine, prostaglandin D₂, and leukotrienes LTC₄, D₄, and E₄, and also by reflex neural pathways.
- The late-phase reaction is inflammatory in nature.
- Inflammatory mediators stimulate epithelial cells to produce chemokines (including eotaxin, a potent chemoattractant and activator of eosinophils) that promote the recruitment of TH2 cells, eosinophils, and other leukocytes, thus amplifying an inflammatory reaction that is initiated by resident immune cells.

Ctd...

Initial sensitization to inhaled antigens which stimulate induction of TH2 type cells that release cytokines such as IL 4 & 5

These cytokines promote IgE production by B cells, growth of mast cells, growth & activation of eosinophils

Subsequent IgE mediated reaction to inhaled allergens elicits an acute response & late phase reactions

Sub epithelial vagal (parasympathetic) stimulation provokes broncho constriction

Ctd...

Sub epithelial vagal (parasympathetic) stimulation provokes broncho constriction

Acute or immediate response consists of broncho constriction, edema, mucus secretion & hypotension

Expose of pre sensitized IgE coated mast cells to antigen stimulates release of chemical mediators

Ctd...

Mast cells also release cytokines that cause the influx of other leukocytes & set the stage of the **late phase reaction**

(4-8hrs later)

Mediators can also be produced by other cells

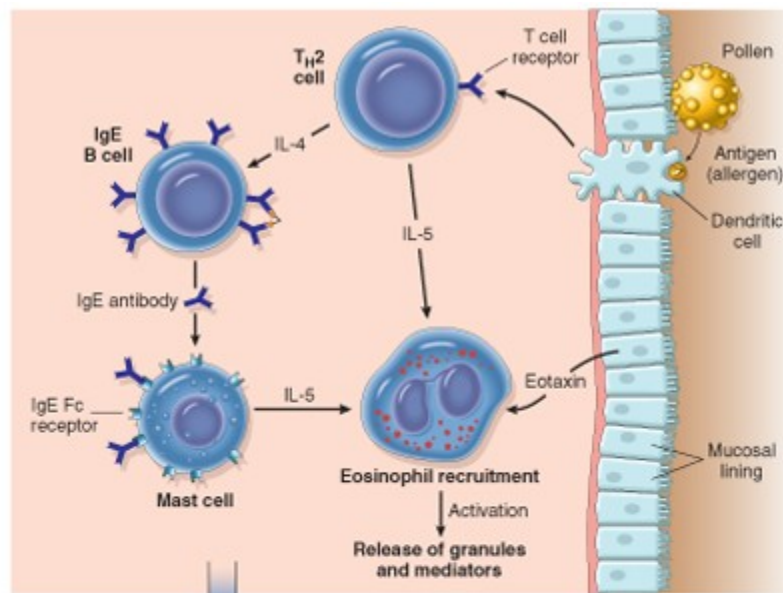
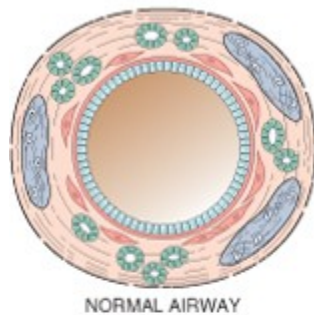
Inflammatory cells already present in asthmatics

Vascular endothelium

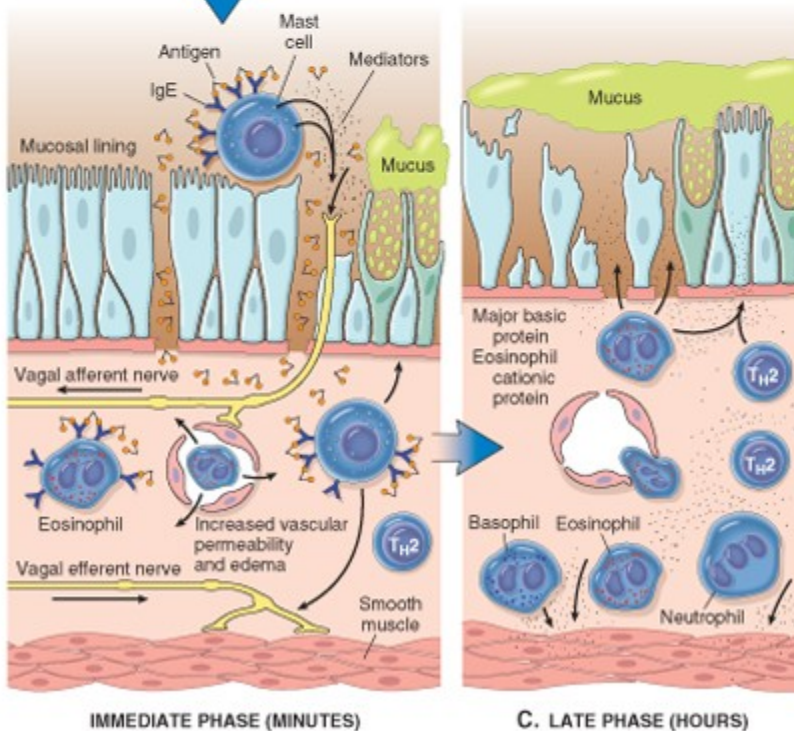
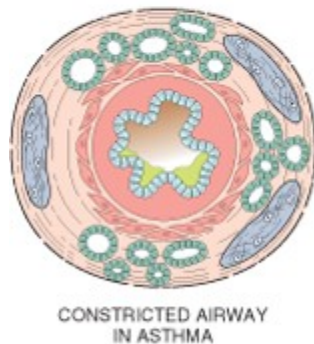
Airway epithelial cells (eotaxin)

MBP (major basic protein) causes epithelial cell damage & airway constriction

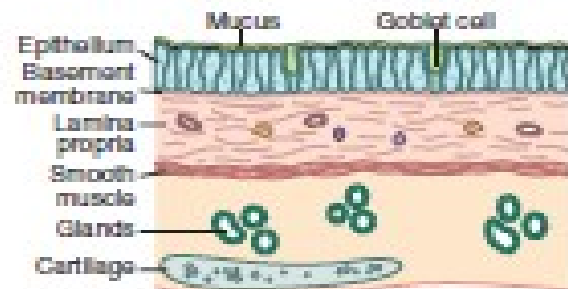
A. SENSITIZATION TO ALLERGEN



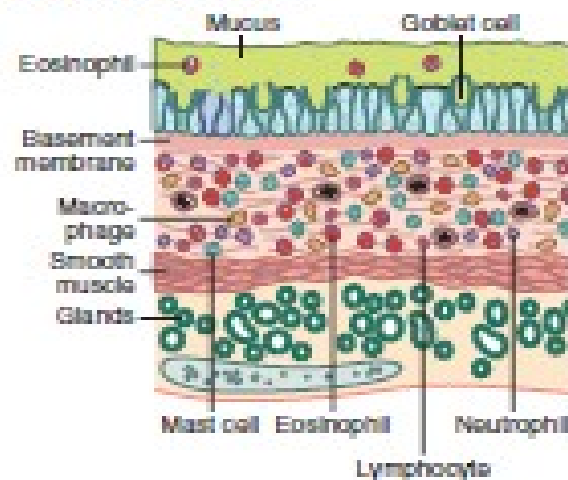
B. ALLERGEN-TRIGGERED ASTHMA



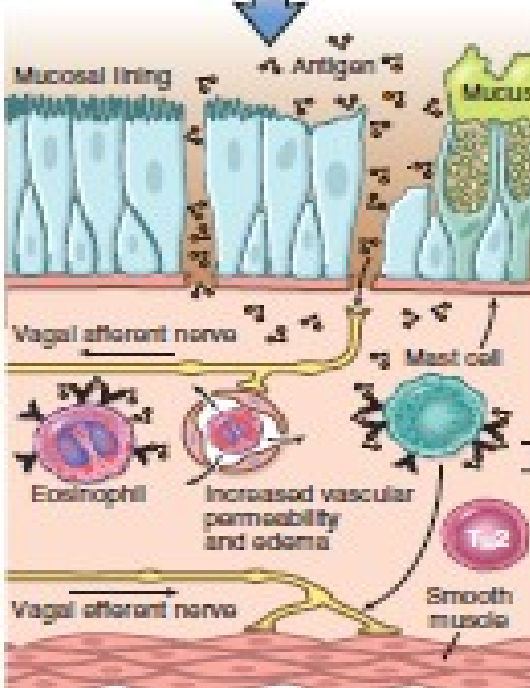
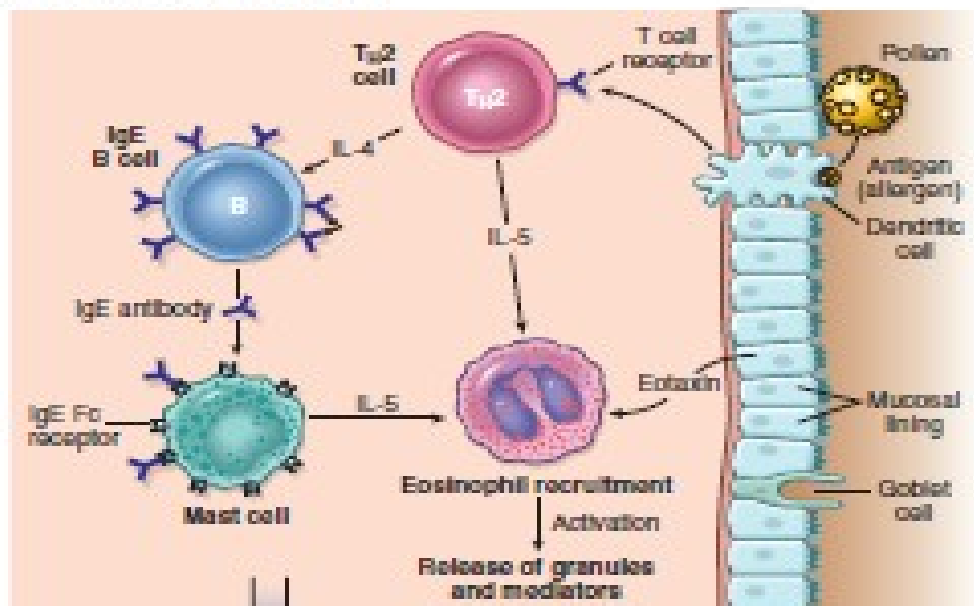
A NORMAL AIRWAY



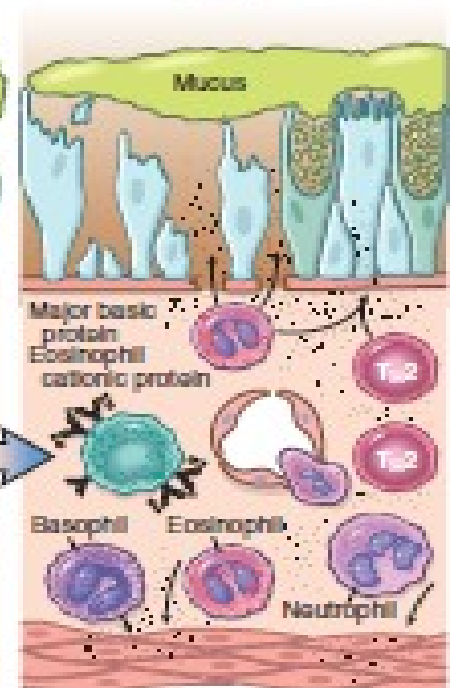
B AIRWAY IN ASTHMA



C TRIGGERING OF ASTHMA



D IMMEDIATE PHASE (MINUTES)



E LATE PHASE (HOURS)

Non-Atopic Asthma

Most frequently triggered by viral respiratory infection

Positive family history is uncommon

Serum IgE level is normal, Skin test is negative

No associated allergens

Virus induced inflammation of the respiratory mucosa lowers the threshold of the sub epithelial vagal receptors to irritants

Inhaled air pollutant may also contribute

Other forms

Drug induced Asthma

Aspirin – tipping the balance toward broncho constrictor leukotrienes

Occupational asthma

Fumes ,dusts

Morphology

Occlusion of bronchi and bronchioles by mucus plug

Mucus plug contain whorls of shed epithelium - **curschmann spirals**

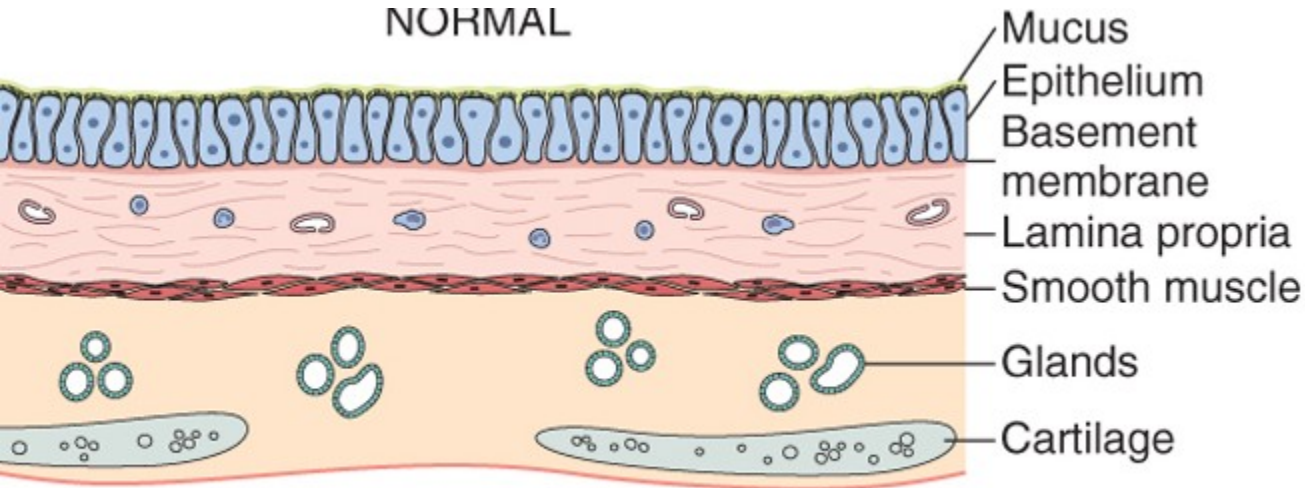
Eosinophils and **charcot leyden crystals**

(Crystalloid made up of eosinophil membrane protein)

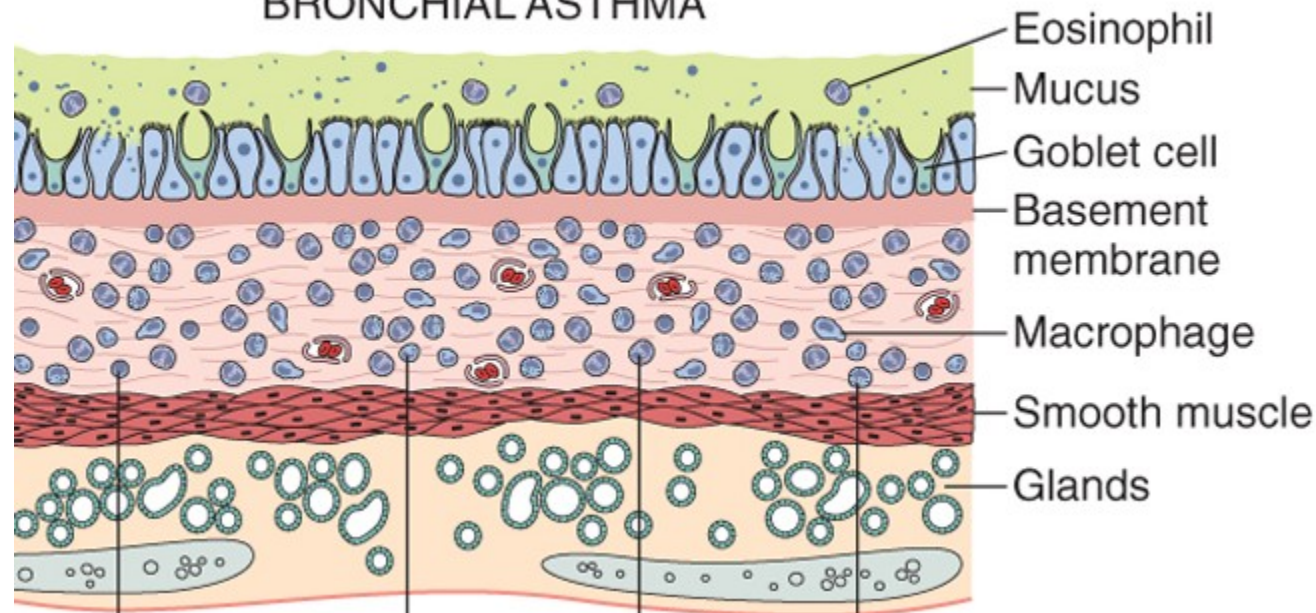
Airway remodeling

- Is characteristic finding of asthma
- Thickening of the basement membrane of the bronchial epithelium
- Edema and inflammatory infiltrate in the bronchial walls, with prominence of eosinophils and mast cells
- Increase in size of the submucosal glands
- Hypertrophy of the bronchial muscle wall

NORMAL



BRONCHIAL ASTHMA



Clinical course

An attack of asthma is characterized by severe dyspnea and wheezing due to bronchoconstriction and mucus plugging, which leads to trapping of air in distal airspaces and progressive hyperinflation of the lungs.

In the usual case, attacks last from 1 to several hours and subside either spontaneously or with therapy.

Intervals between attacks are characteristically free from overt respiratory difficulties, but persistent, subtle deficits can be detected by pulmonary function tests.

Occasionally a severe paroxysm occurs that does not respond to therapy and persists for days and even weeks (status asthmaticus).

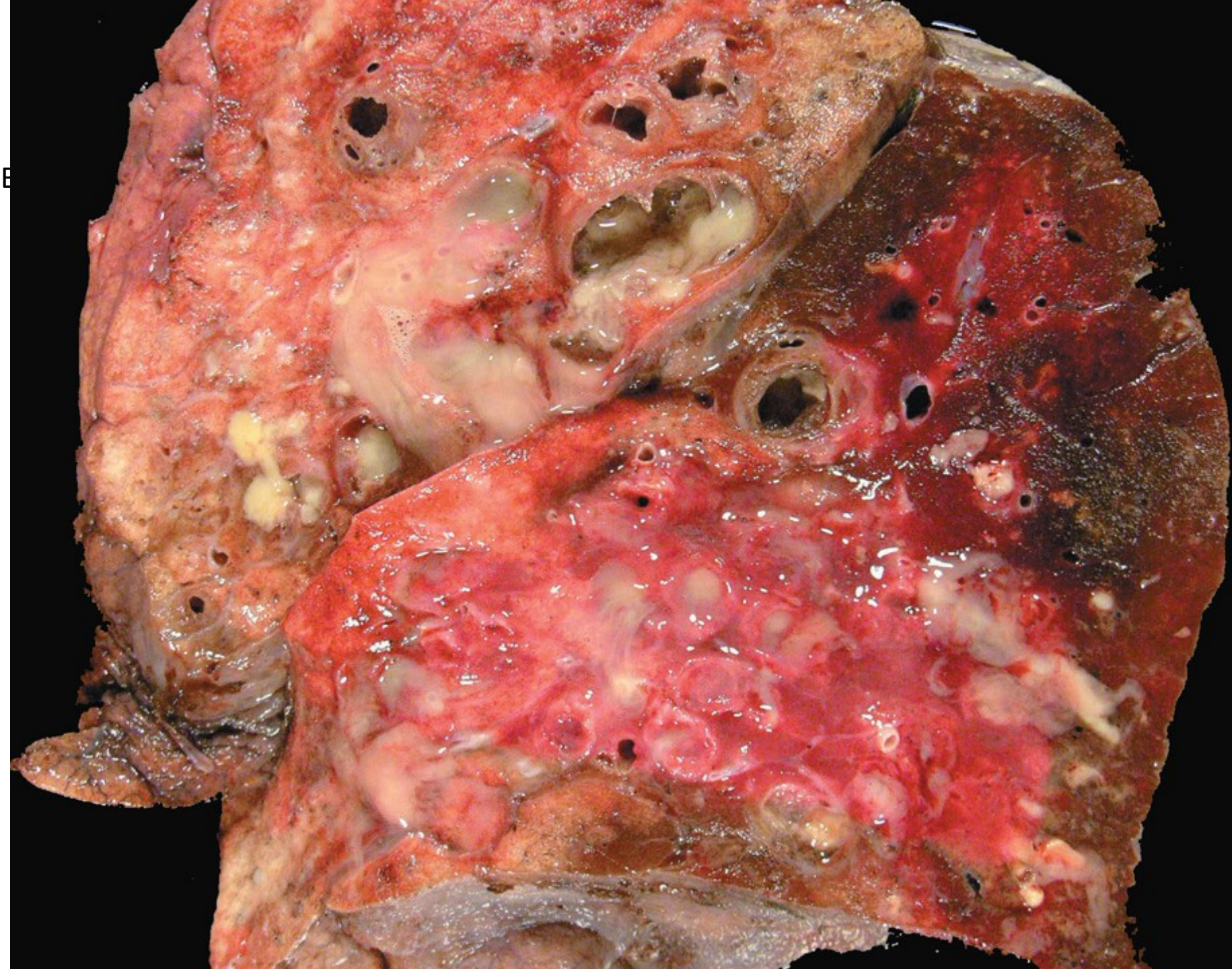
The associated hypercapnia, acidosis, and severe hypoxia may be fatal, although in most cases the condition is more disabling than lethal.

- Standard therapies include anti-inflammatory drugs, particularly glucocorticoids and bronchodilators such as beta-adrenergic drugs and leukotriene inhibitors

Bronchiectasis

A disease characterized by permanent dilation of bronchi and bronchioles caused by destruction of the muscle and elastic tissue resulting from or associated with chronic necrotizing infection manifested by cough, fever, & copious amounts of foul smelling position dependent, purulent sputum

It is not a primary disorder, as it always occurs secondary to persistent infection or obstruction caused by a variety of conditions.



ETIOLOGY and PATHOGENESIS

Obstruction & infection are major influences

Pooling of secretions distal to obstruction ,& there is inflammation of the airway

Severe infections of the airway lead to inflammation, often with necrosis, fibrosis, and eventually dilation of the airways

In primary ciliary dyskinesia ,poorly functioning cilia contribute to the retention of secretions & recurrent infection that in turn gives bronchiectasis

ctd

Bronchiectasis develops in association with
Congenital or hereditary conditions including cystic
fibrosis, primary ciliary dyskinesia , kartagner
syndrome, immunodeficiency, intralobular
sequestration of lung

Post infectious conditions- bacterial, viral, & fungal
infections

Bronchial obstruction – tumors, foreign bodies

Morphology

Usually affects the lower lobes

More severe in the more distal bronchi & bronchioles

The airways are dilated

Cut surface of the lung reveals dilated bronchi appearing as cysts filled mucopurulent secretions

Active case- intense acute & chronic inflammatory exudation associated with desquamation of lining epithelium and areas of ulceration

Chronic cases-bronchial and peribronchial wall fibrosis

Ctd...

The airways are dilated sometimes four times than the normal size

Long tube like enlargement is called cylindrical bronchiectasis

Saccular distension is the other variant of bronchiectasis



Clinical course

Sever, persistent cough

Expectoration of foul smelling, sometimes bloody sputum

Dyspnea, orthopnea, Fever

Mixed flora can be cultured

Complications

Cor pulmonale

Metastatic brain abscess

amyloidosis

Disorders Associated With Airflow Obstruction: The Spectrum of Chronic Obstructive Pulmonary Disease

Clinical Entity	Anatomic Site	Major Pathologic Changes	Etiology	Signs/Symptoms
Chronic bronchitis	Bronchus	Mucous gland hypertrophy and hyperplasia, hypersecretion	Tobacco smoke, air pollutants	Cough, sputum production
Bronchiectasis	Bronchus	Airway dilation and scarring	Persistent or severe infections	Cough, purulent sputum, fever
Asthma	Bronchus	Smooth muscle hypertrophy and hyperplasia, excessive mucus, inflammation	Immunologic or undefined causes	Episodic wheezing, cough, dyspnea
Emphysema	Acinus	Air space enlargement, wall destruction	Tobacco smoke	Dyspnea
Small airway disease, bronchiolitis*	Bronchiole	Inflammatory scarring, partial obliteration of bronchioles	Tobacco smoke, air pollutants	Cough, dyspnea

Acute restrictive disease

ARDS

Synonyms – shock lung, diffuse alveolar damage, acute alveolar injury, acute lung injury

Respiratory failure occurring within 1 week of a known clinical insult

Caused by diffuse alveolar capillary damage

Rapid onset of severe life threatening respiratory insufficiency, cyanosis, severe arterial hypoxemia refractory to oxygen therapy & may progress to multisystem organ failure

- Severe ARDS is characterized by rapid onset of life threatening respiratory insufficiency, cyanosis, and severe arterial hypoxemia that is refractory to oxygen therapy.
- The histologic manifestation of ARDS in the lungs is known as diffuse alveolar damage (DAD).
- ARDS may occur in a multitude of clinical settings and is associated with primary pulmonary diseases and severe systemic inflammatory disorders such as sepsis.
- The most frequent triggers of ARDS are pneumonia (35%–45%) and sepsis (30%–35%), followed by aspiration, trauma (including brain injury, abdominal surgery, and multiple fractures), pancreatitis, and transfusion reactions.
- ARDS should not be confused with respiratory distress syndrome of the newborn; which is caused by a deficiency of surfactant caused by prematurity.

Morphology

- In the acute phase of ARDS, the lungs are dark red, firm, airless, and heavy.
- Microscopic examination reveals capillary congestion, necrosis of alveolar epithelial cells, interstitial and intraalveolar edema and hemorrhage, and (particularly with sepsis) collections of neutrophils in capillaries.
- The most characteristic finding is the presence of hyaline membranes, particularly lining the distended alveolar ducts

- Such membranes consist of fibrin-rich edema fluid admixed with remnants of necrotic epithelial cells.
- Overall, the picture is remarkably similar to that seen in respiratory distress syndrome of the newborn
- In the organizing stage, type II pneumocytes proliferate vigorously in an attempt to regenerate the alveolar lining.
- Resolution is unusual; more commonly, the fibrin-rich exudates organize into intra alveolar fibrosis.
- Marked thickening of the alveolar septa ensues due to proliferation of interstitial cells and deposition of collagen.

Ctd...

Numerous & diverse conditions are associated with ARDS but the commonest four are

- ❖ Sepsis
- ❖ Diffuse pulmonary infections
- ❖ Gastric aspiration
- ❖ Mechanical injury including head injury

Pathogenesis

Central causation is diffuse damage to the alveolar capillary walls

Alveolar capillary damage causes increased vascular permeability

Acute lung injury occurs as a result of cellular events initiated by inflammatory stimuli

Following an acute insult, pulmonary macrophages synthesize IL-8

Ctd...

- Release of mediators lead to activations of neutrophils
- Activated neutrophils releases of oxidases, proteases, PAF, leukotrienes that cause tissue damage
- Exudate & diffuse tissue destruction results in organization, scarring, producing chronic lung diseases
- Transudate of cardiogenic pulmonary edema usually resolves

- The destructive forces unleashed by neutrophils can be counteracted by an array of endogenous anti-proteases and anti-oxidants that are upregulated by proinflammatory cytokines.
- It is the balance between the destructive and protective factors that determines the degree of tissue injury and clinical severity of the ARDS.

Morphology

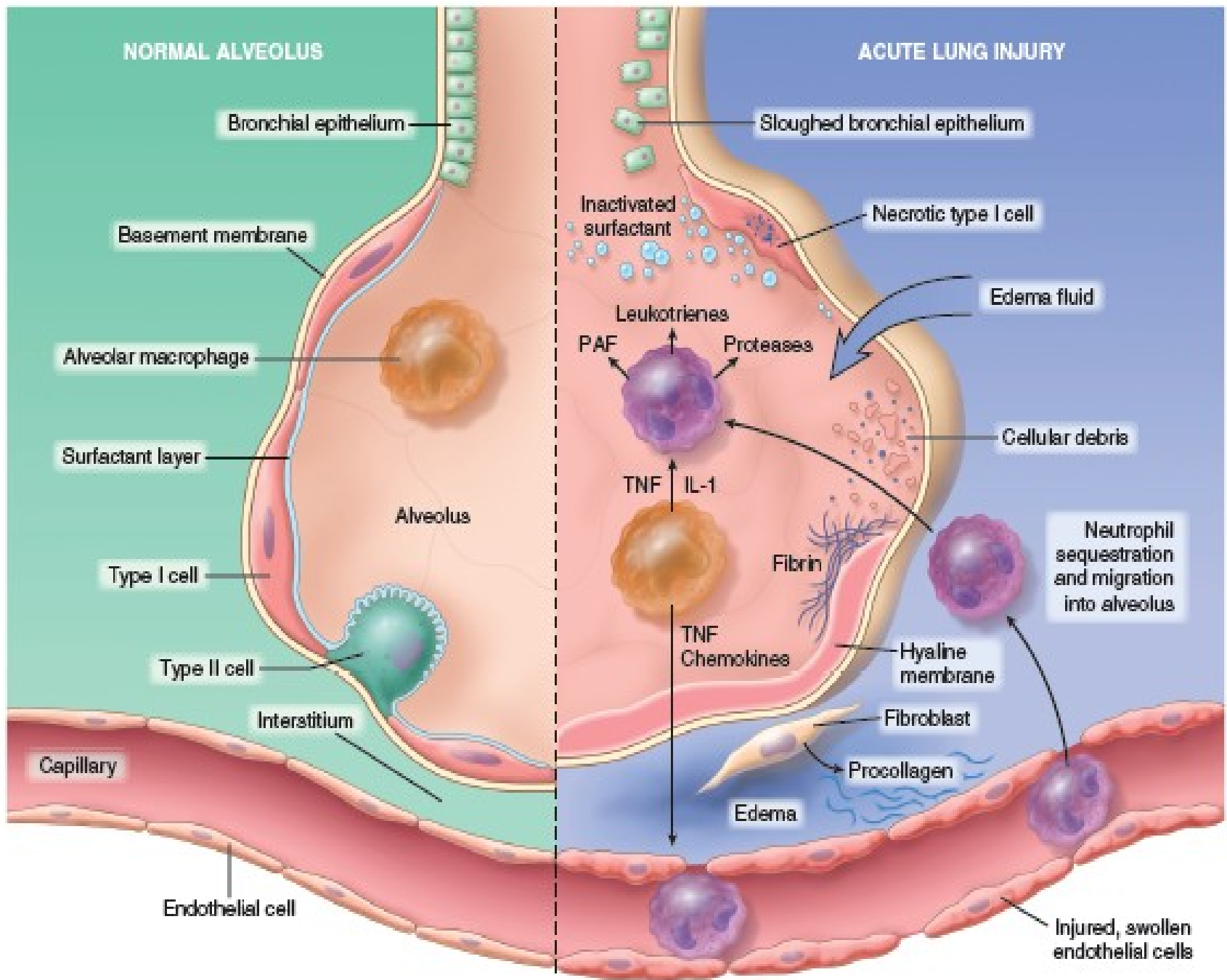
Lungs are heavy, firm, red & boggy

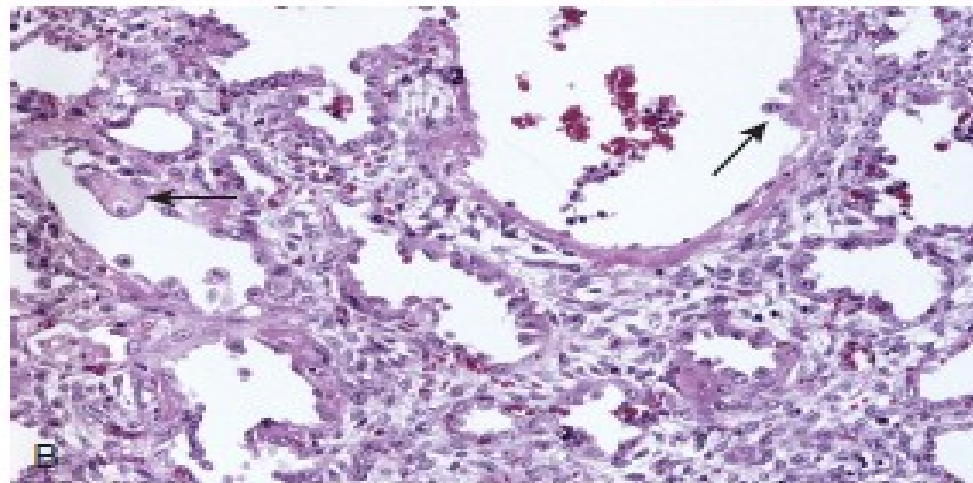
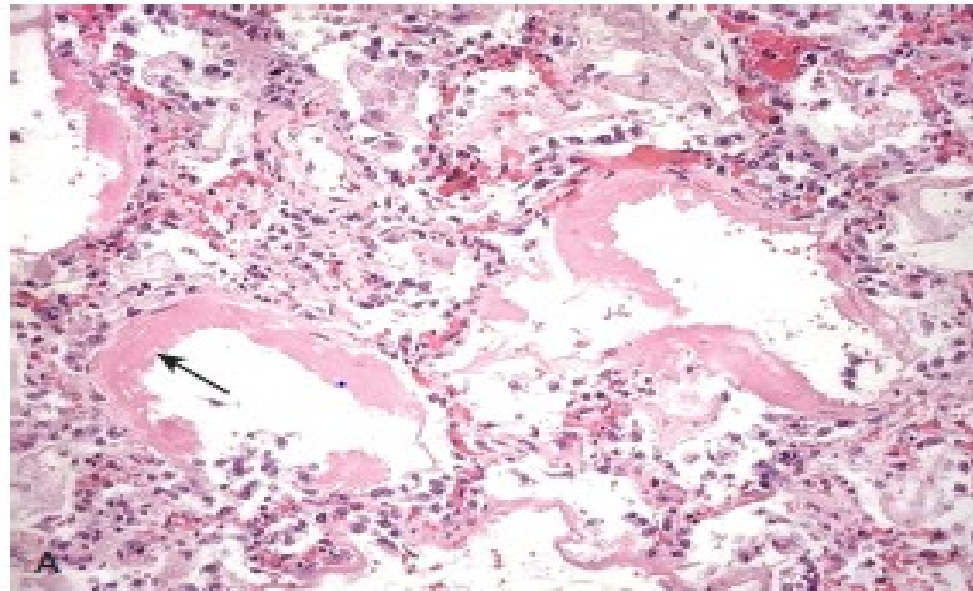
Congestion, interstitial & intra-alveolar edema,
inflammation & fibrin deposition

The alveolar walls become lined with waxy hyaline
membranes

Resolution is unusual, more commonly intra-alveolar
fibrosis

Thickened alveolar septa caused by proliferation of
interstitial cells





Clinical course

Profound dyspnea & tachypnea herald ARDS

Cyanosis, hypoxemia, respiratory failure

Hypoxemia becomes unresponsive to oxygen therapy

Respiratory acidosis

Mortality rate is about 60%

X- ray may be normal initially, later there will be diffuse bilateral infiltrates

C/F

- In 85% of cases, it develops within 72 hours of the initial insult.
- The overall hospital mortality rate is 38.5% (27%, 32%, and 45% for mild, moderate, and severe ARDS, respectively).
- Predictors of poor prognosis include advanced age, bacteremia (sepsis), and the development of multiorgan failure.
- Most patients who survive the acute insult recover normal respiratory function within 6 to 12 months, but the rest develop diffuse interstitial fibrosis leading to chronic respiratory insufficiency.

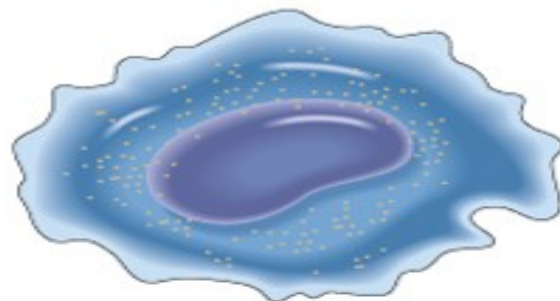
CHRONIC INTERSTITIAL(RESTRICTIVE, INFILTRATIVE) LUNG DISEASES

- Chronic interstitial diseases are heterogeneous group of disorders characterized by bilateral, often patchy, pulmonary fibrosis mainly affecting the walls of the alveoli
- Many of the entities in this group are of unknown cause and pathogenesis; some have an intraalveolar and an interstitial component

Ctd...

- The hallmark of these disorders is reduced compliance (i.e., more pressure is required to expand the lungs because they are stiff), which in turn necessitates increased effort of breathing (dyspnea).
 - Patients have dyspnea, tachypnea & respiratory crackles, cyanosis with out wheezing or other evidence of airway obstruction
 - CXR shows diffuse infiltration by small nodules, irregular lines or ground glass shadows
- With progression, patients may develop **respiratory failure, pulmonary hypertension, and cor pulmonale**

LUNG INJURY
Inhaled agents, dusts, blood-borne toxins,
unknown antigens



ACTIVATED MACROPHAGE

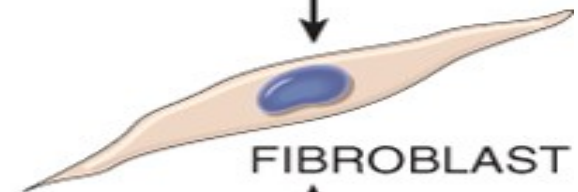
Recruitment
of neutrophils



Oxidants
Proteases

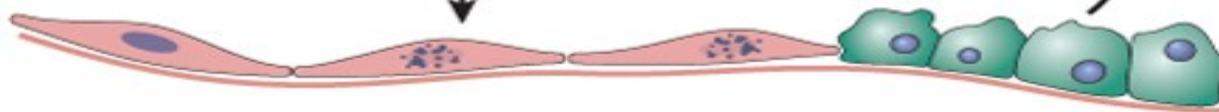
Injury to type I
pneumocytes

Fibrogenic and
chemotactic cytokines



FIBROBLAST

Fibrogenic and
chemotactic cytokines



Hypertrophy and hyperplasia
of type II pneumocytes

Major categories of chronic interstitial lung disease

Fibrosing

Usual interstitial pneumonia (idiopathic pulmonary fibrosis)
Nonspecific interstitial pneumonia
Cryptogenic organizing pneumonia
Collagen vascular disease-associated
Pneumoconiosis
Therapy-associated (drugs, radiation)

Granulomatous

Sarcoidosis
Hypersensitivity pneumonia

Eosinophilic

Loeffler syndrome
Drug allergy–related
Idiopathic chronic eosinophilic pneumonia

Smoking-Related

Desquamative interstitial pneumonia
Respiratory bronchiolitis

Pneumoconiosis

Used to describe the non neoplastic lung rxn to inhalation of mineral dusts, organic particulates, chemical fumes & vapors

Agent	Disease	Exposure
Coal dust	Simple coal worker's pneumoconiosis: macules and nodules Complicated coal worker's pneumoconiosis: PMF	Coal mining
Silica	Silicosis	Sandblasting, quarrying, mining, stone cutting, foundry work, ceramics
Asbestos	Asbestosis, pleural effusions, pleural plaques, or diffuse fibrosis; mesothelioma; carcinoma of the lung and larynx	Mining, milling, and fabrication of ores and materials; installation and removal of insulation

PMF, Progressive massive fibrosis.

Pathogenesis

Regardless of the cause the earliest manifestation of interstitial disease is alveolitis i.e accumulation of inflammatory & immune effector cells within the alveolar walls & spaces.

Leukocyte accumulation distorts alveolar structure & release of mediators resulting in fibrosis

General pathogenesis

The development of pneumoconiosis depends on

- dust retained in the lungs & airways
- The size, shape of the particles
- particle solubility & physiochemical reactivity
- Additional effects of other irritants eg. Smoking

1. Coal workers' pneumoconiosis

There are three spectrum of this disease

I. Asymptomatic anthracosis

II. Simple coal workers' pneumoconiosis

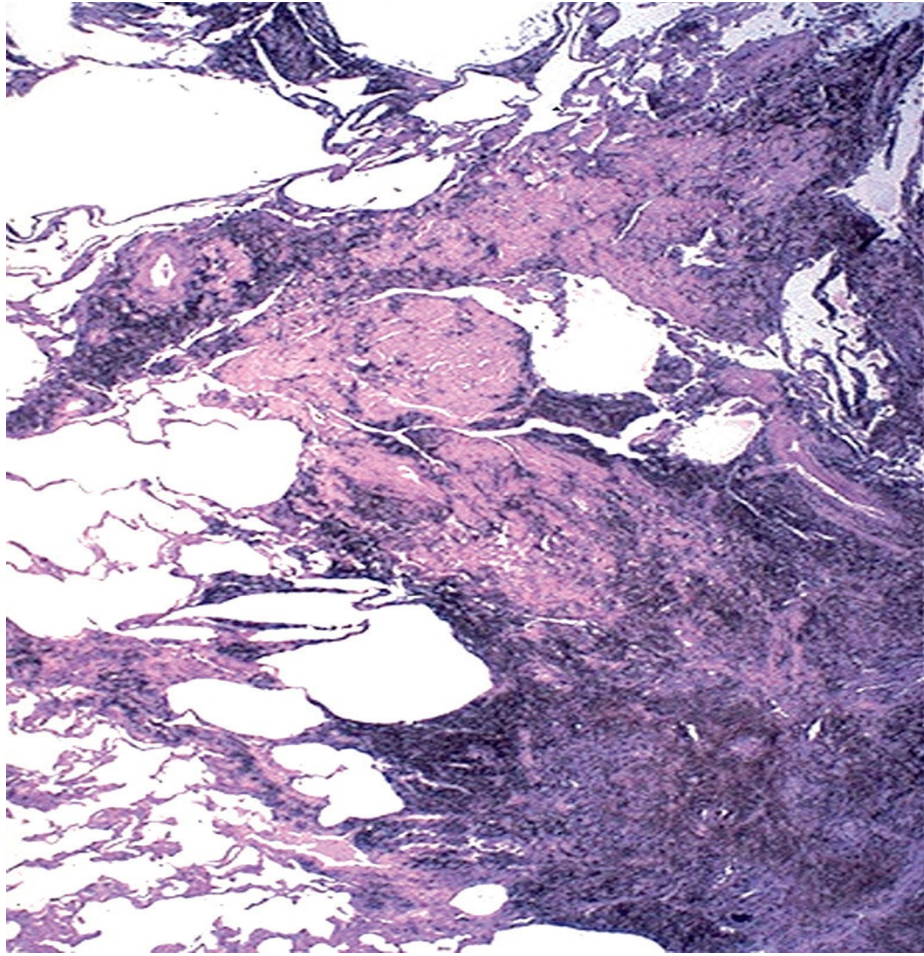
III. Complicated coal workers'

Morphology

Anthracosis – innocuous, coal induced pulmonary lesion, commonly seen in all urban dwellers & tobacco smokers & the inhaled carbon accumulate in connective tissue.

Simple CWP – coal macules containing carbon laden macrophages 1-2mm in diameter, coal nodules, upper lobe heavily involved

Complicated CWP – generally requires many years to develop , intense black scars, dense collagen & pigment with areas of necrosis



Clinical course

CWP benign disease with little decrement in lung function.

Its exposure is during coal mining

< 10% develop PMF leading to pulmonary dysfunction, pulmonary HTN & cor pulmonale

Increased risk of chronic bronchitis & emphysema

CWP doesn't predispose to cancer independent of smoking

2. Silicosis

Caused by inhalation of crystalline silicon dioxide (silica)
Exposure sand blasting, hard rock mining, stone cutting
There are two forms of silica-crystalline form which is the most fibrogenic & amorphous form
crystalline forms (including quartz, cristobalite, and tridymite) are by far the most toxic and fibrogenic.
Of these, quartz is most commonly implicated in silicosis.
After inhalation the particles will interact with epithelial cells & macrophages

Ctd...

Some of the cytokines include IL-1, TNF, fibronectin, lipid mediators, oxygen derived free radical & fibrogenic cytokines

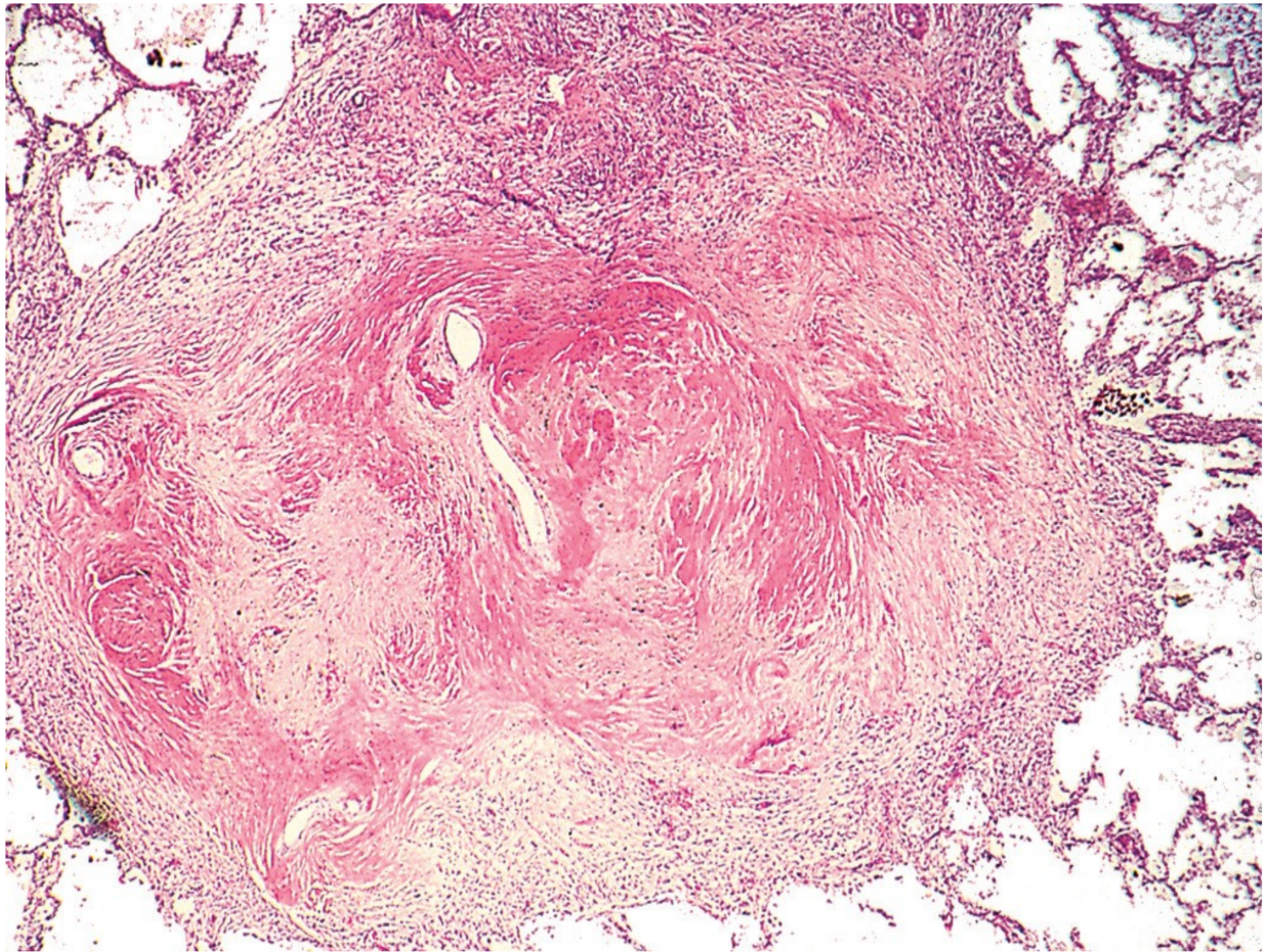
Morphology

Hard collagenous scar

Some nodules undergo central softening & cavitations

CXR egg shell calcification

Examination by polarized microscopy birefringent silica



Clinical feature

Most prevalent chronic occupational disease in the world

Occurs after decades of exposure & cause nodular fibrosing pneumoconiosis

It increases susceptibility to TB

Most patients do not develop dyspnea until late in the course

Its relation with cancer is controversial

3. Asbestosis related disease

Its exposure is linked to

- Pleural effusion or fibrosis

- Asbestosis

- Lung cancer (5x)

- Mesothelioma

- Laryngeal or colonic cancer

It causes diffuse interstitial disease unlike silicosis which causes nodular fibrosing disease

Asbestosis can act as tumor initiator or promoter

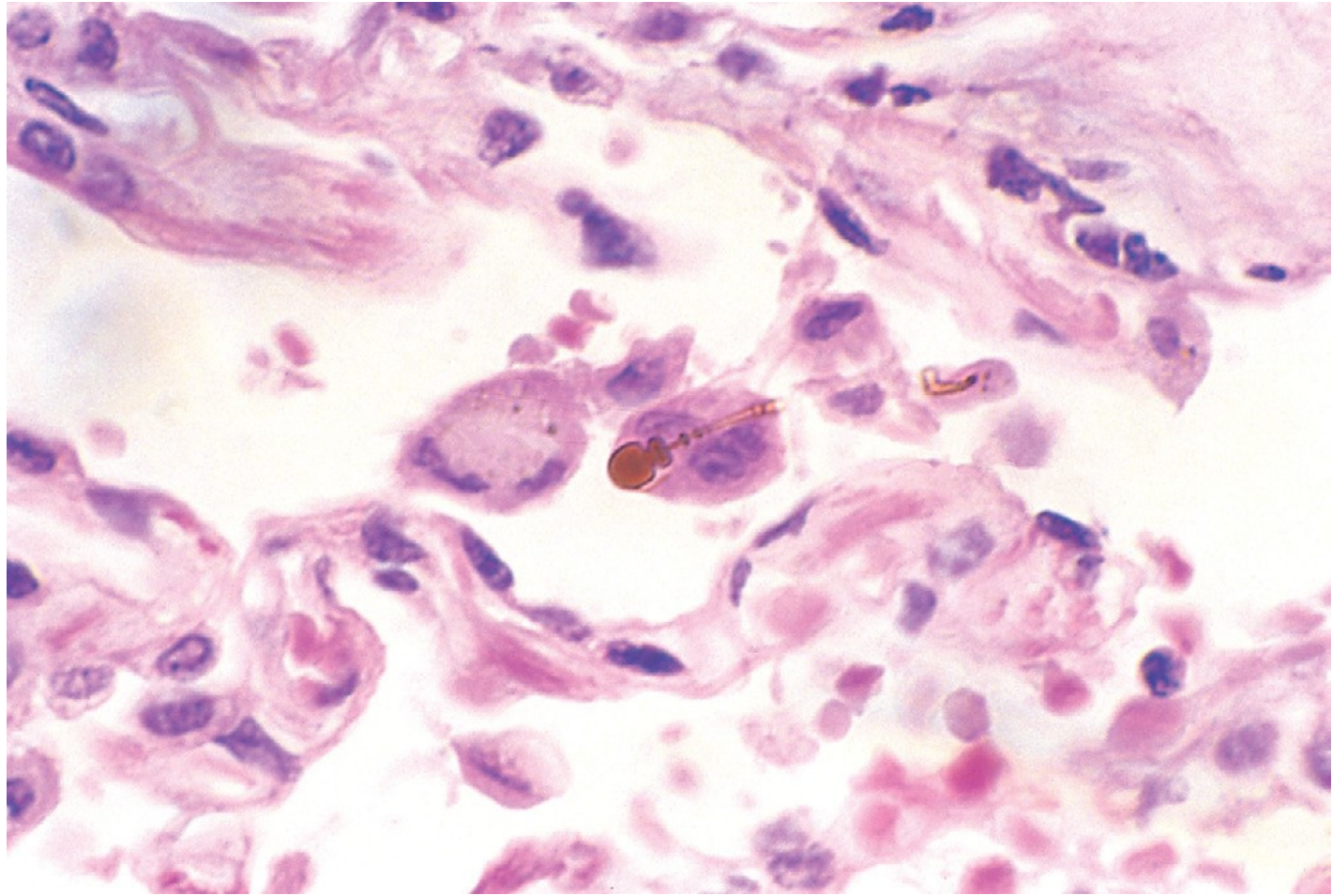
Morphology

- It is difficult to differentiate from other diffuse interstitial disease unless there is asbestos bodies

Asbestosis is marked by diffuse pulmonary interstitial fibrosis, characterized by the presence of asbestos bodies, which are seen as golden brown, fusiform or beaded rods with a translucent center.

They consist of asbestos fibers coated with an iron-containing proteinaceous material

- Fibrosis begins around the bronchioles & alveolar ducts
- Unlike other forms of pneumoconiosis it affects the lower lobes
- Pleural plaques are the most common manifestations of asbestosis
- May be complicated by pulmonary HTn & cor pulmonale



Idiopathic pulmonary fibrosis

- Also called cryptogenic fibrosing alveolitis

Patchy, progressive bilateral interstitial fibrosis

- It refers to a clinicopathologic syndrome with a characteristic radiologic, pathologic & clinical features

The radiologic and histologic pattern of fibrosis is referred to as usual interstitial pneumonia (UIP), which is required for the diagnosis of IPF

- Histological features of usually interstitial pneumonia is required for its diagnosis

Pathogenesis

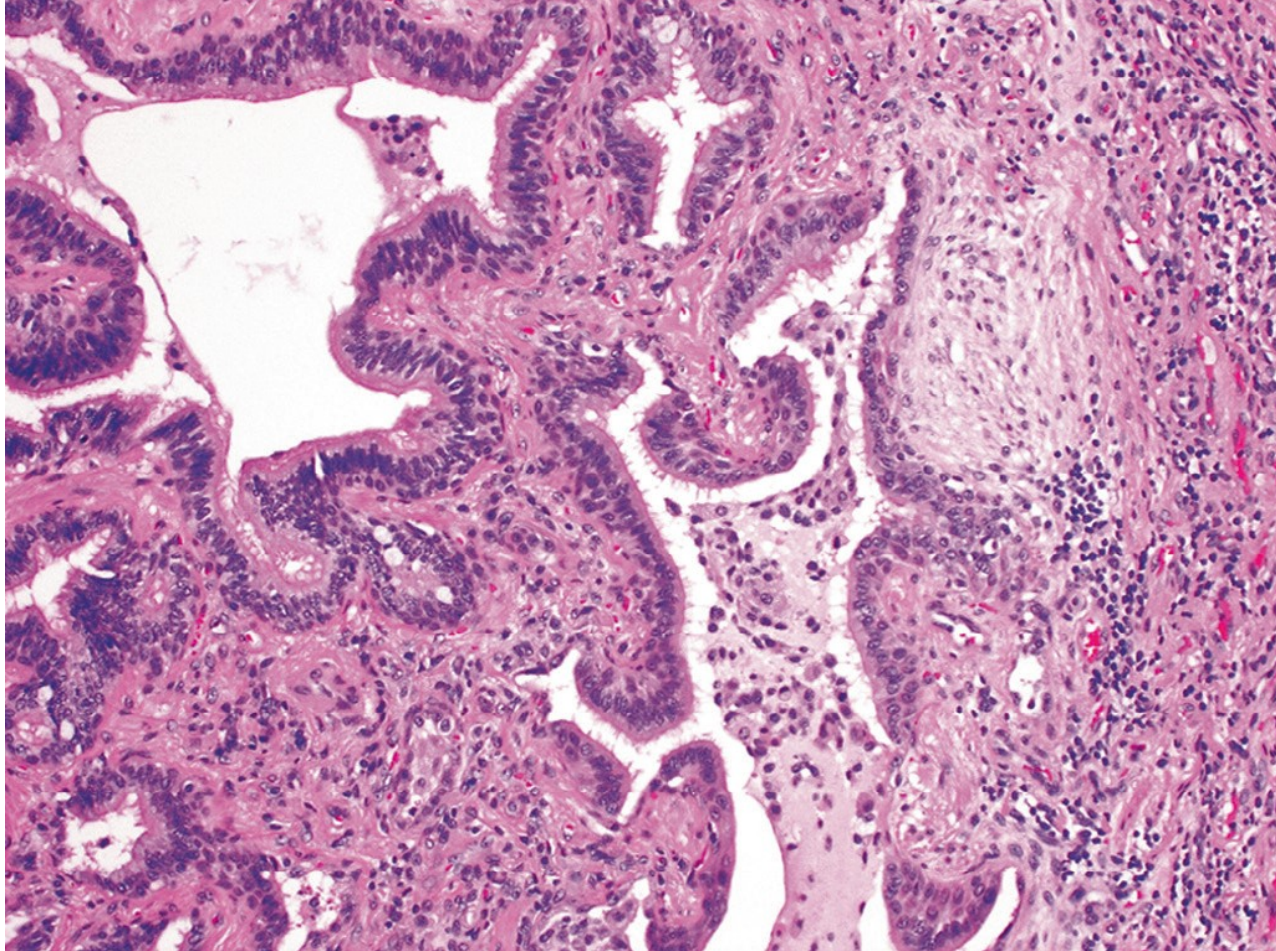
- Unknown etiology
- It is caused by repeated cycles of acute lung injury (alveolitis) by some undefined agents
- Wound healing at these sites give rise to exuberant fibrosis
- The inflammatory response is mediated by TH2 cell types
- Genetic & environmental factors modify the phenomenon

Morphology

- Pleural surface grossly cobblestoned owing to retraction of scars along the interlobular septa
- Microscopically the hallmark UIP is **patchy interstitial fibrosis** & the earliest manifestation is **fibroblastic foci**

Ctd...

- Honey comb fibrosis dense, causes collapse of alveolar walls & formation of cystic spaces lined by hyperplastic type II pneumocytes or bronchial epithelium.
- Infiltration of lymphocytes
- Foci of squamous metaplasia & smooth muscle hyperplasia
- Secondary pulmonary HTn due to intimal fibrosis & medial thickening of the pulmonary arteries



Clinical course

- Age 40-70 yrs (>60yrs)
- Dyspnea
- Dry cough
- Hypoxemia
- Cyanosis
- Clubbing
- Mean survival is 3 yrs
- Lung transplantation is the only definitive therapy

Non specific interstitial pneumonia

- Etiology unknown
- Lung biopsies failed to show diagnostic feature of any of the DILD, a "wastebasket" type of diagnosis
- Better prognosis than UIP (usual interstitial pneumonia)
- Morphologically, it has cellular & fibrosing patterns
- Cellular components include chronic inflammatory cells, has good outcome
- Diffuse & patchy fibrosis
- Younger than UIP

Drug- and Radiation-Induced Pulmonary Disease

- Drugs can cause a variety of acute and chronic alterations in respiratory structure and function.
- For example, bleomycin, an anti-cancer agent, causes pneumonitis and interstitial fibrosis as a result of direct toxicity of the drug and by stimulating the influx of inflammatory cells into the alveoli.
- Amiodarone, an anti-arrhythmic agent, also is associated with risk for pneumonitis and fibrosis.
- Radiation pneumonitis is a well-known complication of irradiation of pulmonary and other thoracic tumors.
- Acute radiation pneumonitis, which typically occurs 1 to 6 months after therapy in as many as 20% of the patients, is manifested by fever, dyspnea out of proportion to the volume of irradiated lung, pleural effusion, and pulmonary infiltrates in the irradiated lung bed.
- These signs and symptoms may resolve with corticosteroid therapy or progress to chronic radiation pneumonitis, associated with pulmonary fibrosis.

Pulmonary involvement in collagen vascular diseases

- In **rheumatoid arthritis** pulmonary involvement is common in one of the four forms
 - I. Chronic pleuritis with or without effusion
 - II. Diffuse interstitial pneumonitis or fibrosis
 - III. Intrapulmonary rheumatoid nodules
 - IV. Pulmonary hypertension

Granulomatous Diseases

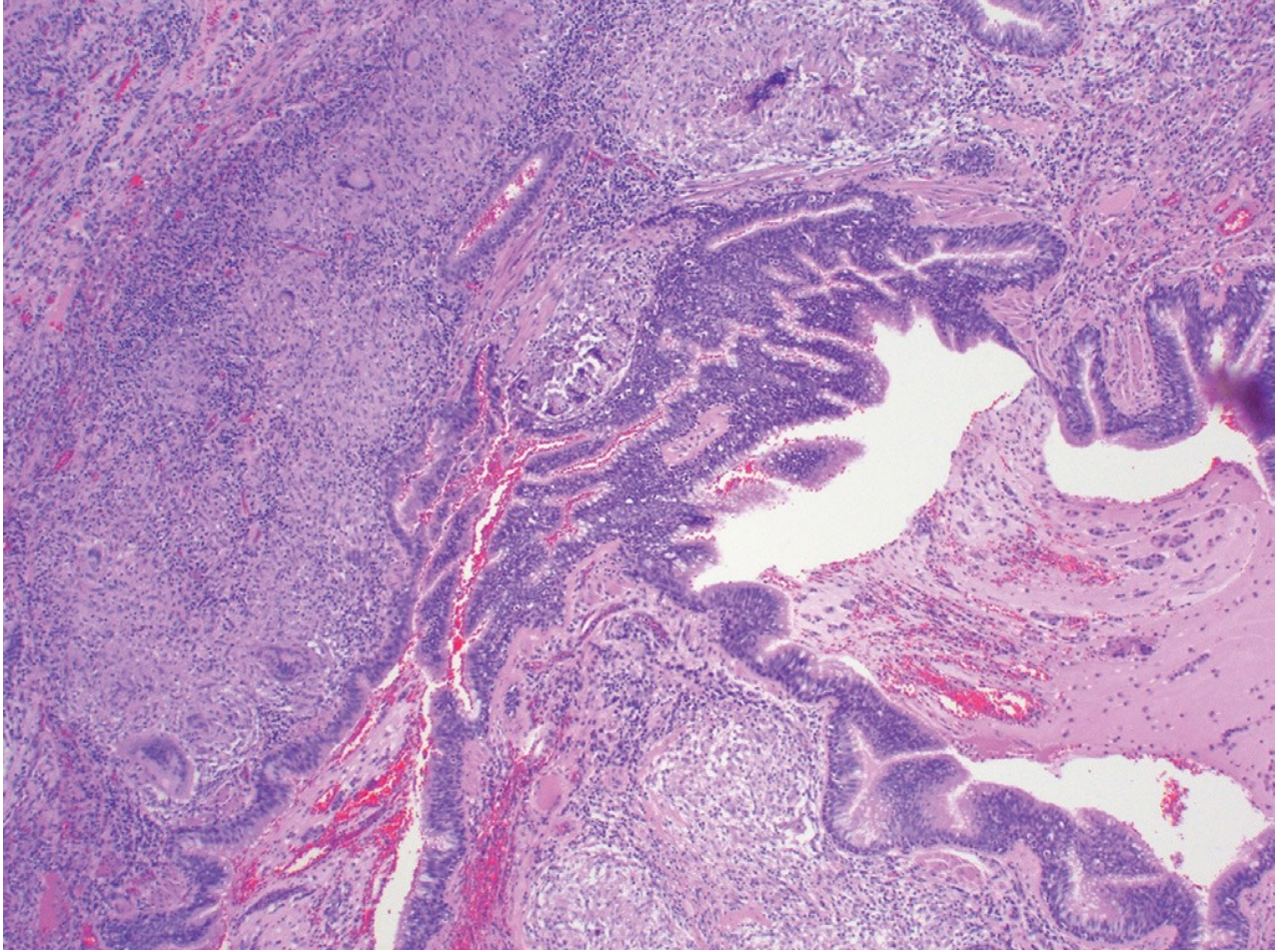
Sarcoidosis

- sarcoidosis is a multisystem disease of unknown etiology characterized by noncaseating granulomas in many tissues and organs
- bilateral hilar lymphadenopathy or lung involvement (or both), visible on chest radiographs, is the major presenting manifestation in most cases.
- Eye and skin involvement each occurs in about 25% of cases and may occasionally be the presenting feature of the disease.

- Common after the age of 40 years
- Unknown etiology, disorder of immune regulation in genetically predisposed individuals exposed to certain environmental agents.
- T-cell mediated reaction
- Several putative antigen have been proposed (viruses, mycobacteria, *Borrelia*, pollen), but thus far *there is no unequivocal evidence to suggest that sarcoidosis is caused by an infectious agent.*

Morphology: regardless of the organ involved

- **Non caseating epitheloid granuloma**
- **Asteroid bodies**(inclusions in multinucleated giant cells)
- **Schaumann bodies: laminated** concretions composed of calcium and proteins



Hypersensitivity pneumonitis

- **Immunologically mediated disorder affecting airways and interstitium.**
- is an immunologically mediated response to an extrinsic antigen that involves both immune-complex and delayed-type hypersensitivity reactions

Pulmonary Eosinophilia

- These diverse diseases generally are of immunologic origin, but the etiology is not understood.
- **Pulmonary eosinophilia is divided into the following categories:**
- **Acute eosinophilic pneumonia with respiratory failure**
- Characterized by rapid onset of fever, dyspnea, hypoxia, and diffuse pulmonary infiltrates on chest radiographs.
- There is prompt response to corticosteroids.
- **Simple pulmonary eosinophilia (Loeffler syndrome)**
- Characterized by transient pulmonary lesions, eosinophilia in the blood, and a benign clinical course.
- **Tropical eosinophilia**
- caused by infection with microfilariae and helminthic parasites
- **Secondary eosinophilia**
- seen, for example, in association with asthma, drug allergies, and certain forms of vasculitis
- **Idiopathic chronic eosinophilic pneumonia**
- Characterized by aggregates of lymphocytes and eosinophils within the septal walls and the alveolar spaces, typically in the periphery of the lung fields, and accompanied by high fever, night sweats, and dyspnea.
- This is a disease of exclusion, once other causes of pulmonary eosinophilia have been ruled out.

Smoking-Related Interstitial Diseases

- Desquamative interstitial pneumonia (DIP) and respiratory bronchiolitis are two related examples of smoking-associated interstitial lung disease.
- The most striking histologic feature of DIP is the accumulation of large numbers of macrophages containing dusty-brown pigment (smoker's macrophages) in the air spaces

- Respiratory bronchiolitis is a common lesion found in smokers that is characterized by the presence of pigmented intraluminal macrophages akin to those in DIP, but in a “bronchiolocentric” distribution (first- and second-order respiratory bronchioles).
- Mild peri bronchiolar fibrosis also is seen.
- As with DIP, affected patients present with gradual onset of dyspnea and dry cough, and the symptoms recede with smoking cessation.

- Most often it is an occupational disease that results from heightened sensitivity to inhaled antigens such as moldy hay.
- Unlike bronchial asthma, in which bronchi are the focus of immunologically mediated injury, the damage in hypersensitivity pneumonitis occurs at the level of alveoli.
- Hence, it presents as a predominantly restrictive lung disease with decreased diffusion capacity, lung compliance, and total lung volume.